DEPARTMENT OF MATHEMATICS

NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. I Semester (**CSE/IT/AIML/IIOT/MC/MV/ADS/PI**) **Mid Term – II Examination**

Date: Nov. 10, 2025 **Max. Marks**: 15 **Time**: 10:00 am – 10:50 am

Paper: Differential Calculus and Differential Equations (MAIC 101) **All questions carry equal marks**

Instructions: The calculator is not allowed into the examination hall, No extra answer sheets will be provided.

1. If $u = \sin^{-1}\left(\frac{x^3+y^3+z^3}{ax+by+cz}\right)$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$ at $u = 0$.
2. Solve by the method of variation of parameters: $x^2y'' + xy' - y = x^2e^x$.
3. Using the Laplace transform, evaluate the integral: $\int_0^\infty \frac{e^{-t}\sin\sqrt{3}t}{t}\,dt$.